Effects of dietary proteins on plasma lipoprotein levels in normal subjects: interaction with dietary cholesterol.

We have compared the effects of dietary soy protein and casein in diets low in cholesterol (less than 100 mg/d) and in diets enriched in cholesterol (500 mg/d) to examine whether the level of cholesterol intake affects the response of plasma lipoproteins to dietary proteins of plant and animal origin. Normal men and women consumed formula diets containing 20% of calories as soy protein or casein, 27% as fat and 53% as carbohydrate in 2 crossover studies. The dietary periods lasted for 31 days and were separated by a month-long interim period on self-chosen food. Following an initial reduction of plasma total cholesterol and low-density lipoprotein-cholesterol (LDL-C) levels on all diets, the plasma lipid and lipoprotein concentrations stabilized. On low-cholesterol diets the concentration of each of the major lipoprotein classes were similar during the soy and the casein dietary periods. On cholesterol-enriched diets, the concentration of LDL-C stabilized at a 16% lower level on soy protein than on the casein diet (p less than 0.02), while the concentration of high-density lipoprotein-cholesterol (HDL-C) was 16% higher (p less than 0.01). Since the difference in LDL-C (p less than 0.05) and in HDL-C (p less than 0.025) levels on casein and on soy protein diets were significantly greater on the high than on the low cholesterol intake, the findings indicate that the level of dietary cholesterol may determine whether plant and animal dietary proteins have similar or different effects on plasma LDL-C and HDL-C concentrations.